

Rohan Jain

jainrohanj@gmail.com | <https://www.linkedin.com/in/rohan-jain11/> | <https://github.com/rohanjain11>
<https://rohanjain11.github.io/Portfolio/> | +1 720-819-0419 | Vista, CA

EDUCATION

University Of Colorado, Boulder

Master's in Data Science (GPA: 3.9/4.0)

Aug 2024 - May 2026 | Boulder, USA

Mumbai University, Rajiv Gandhi Institute of Technology

Bachelor's of Engineering in Computer Engineering (GPA: 3.7/4)

Jul 2020 - May 2024 | Mumbai, India

SKILLS

Languages & Data: Python, R, SQL (MySQL, PostgreSQL; CTEs, window functions), MongoDB. **Analytics & ML:** pandas, NumPy, scikit-learn; EDA, KPI design, forecasting, clustering, association rules; evaluation (MAE, MSE, CRPS). **Engineering & Platforms:** FastAPI, REST APIs, Git/GitHub, Docker (basics); reproducible run artifacts, validation gates, documentation. **Visualization & BI:** Matplotlib, Plotly, Tableau, Power BI, Excel (PivotTables, XLOOKUP, Power Query). **Additional:** TensorFlow, PyTorch, Keras, LangChain, FAISS, Apache Spark; AWS (Lambda, S3), Azure.

EXPERIENCE

Data Science & Engineering Intern, Nexus Weather & Climate

May 2025 - Present | Boulder, CO

- Upgraded a legacy SVM bias-correction step that was limiting forecast accuracy by tuning and validating a Random Forest replacement, then documenting rollout for production promotion; reduced temperature MAE from 0.73 to 0.39 (47%) and humidity MAE from 4.68 to 2.37 (49%).
- Addressed the need for clearer uncertainty signals by building a probabilistic CNN-LSTM trained with a custom CRPS loss, then packaging simple plots and short summaries for review by non-ML stakeholders; achieved a best reported CRPS of 0.1787 for ensemble calibration.
- Reduced inconsistency in model comparisons by creating a reproducible training and benchmarking workflow with schema, unit, and range checks, then training SVR, RF, tuned RF, GBRT, XGBoost, and MLP under the same evaluation rules; produced consistent artifacts with MAE and MSE tracked against pre-ML baselines.
- Prevented regressions from moving forward by implementing an inference gate that requires post-ML MAE to be lower than the pre-ML baseline before a model is accepted; reduced failed runs by 90% by automatically rejecting updates that did not beat baseline performance.
- Improved training decision-making by running controlled experiments where only the loss function changed (CRPS vs hybrid CRPS+MSE with $\alpha = 0.80$ vs Huberized CRPS) and recording runtime plus delta CRPS outcomes; hybrid ran in 3.65 min vs 4.4 min baseline with mean delta post-CRPS of -0.01 to -0.03, while Huberized showed -0.005 to -0.02 with more stable validation behavior.

Graduate Research Assistant, Kopf Lab, University of Colorado Boulder

Dec 2025 - Present | Boulder, CO

- Improved stable isotope data ingestion by standardizing parsing outputs across vendor formats, enforcing consistent field names, units, and schemas so the same file yields the same structure across platforms and package versions for repeatable analysis.
- Reduced silent data issues by adding fail-fast validation for corruption and edge cases using type, bounds, and structural checks (missing blocks, wrong ordering, truncation), returning clear errors instead of partial or incorrect results during ingestion.
- Decoded legacy Thermo Isodat scan (.scn) files stored via MFC CArchive Serialize by mapping class layouts and field order, then implemented R readers that follow the same read sequence and reconstruct analysis-ready objects with traceable provenance.
- Improved debugging and maintenance by attaching file IDs, format signals, parsing paths, and assumptions to outputs, and building regression tests with golden outputs plus corruption fixtures so updates do not break prior behavior or mask parsing errors.
- Designed a multi-user app hosting setup by separating authentication from app execution, using Keycloak OIDC for login and role/group access control while ShinyProxy runs one container per user session, enabling safer operations and maintainable updates.

PROJECTS

SaferRide (Risk-Aware Bicycle Routing) | Python, FastAPI, SQL, PostgreSQL, PostGIS,

Sep 2025 - Dec 2025

GeoPandas, OSRM, Map UI

- Created SaferRide, a bicycle routing app that requests OSRM routes, offers multiple alternatives, and scores each route using crash density and hazard proximity; built a pipeline to clean city datasets, computed route-level risk with PostGIS spatial joins and GeoPandas, and served routes plus risk metrics via FastAPI with map overlays so riders compare safety versus time tradeoffs.

Denver International Airport Analytics | Python, Power BI, Tableau

Oct 2024

- Integrated ServiceNow and Azure DevOps data into analysis-ready pipelines, defined KPIs and built Power BI and Tableau dashboards with drill-downs for SLA status, queue times, and bottlenecks; authored reusable ETL queries and governance notes and KPI definitions for consistent refresh logic, reduced manual reporting by 40%, and earned 3rd place in a hackathon for dashboard impact.

AI PDF Chatbot | Python, FastAPI, React, LangChain, FAISS

Jan 2025

- Built a PDF question-answering tool for scanned and digital documents by combining OCR, embeddings, and FAISS retrieval behind a FastAPI and React UI; standardized chunking and prompt templates, supported secure uploads and real-time responses, added usage logging to track common queries, and ran labeled spot checks to refine retrieval so recurring research questions returned more relevant passages.

PUBLICATION

- Technical paper: *Yoga Posture Detection and Correction*. [Link](#)